Improving Temporal Consistency in VISTA with Bounding Box LoRA

1. Introduction

VISTA can generate realistic autonomous-driving videos, but object identity and appearance may degrade over many frames, particularly for vehicles and other road agents. This experiment investigates whether explicit object-level conditioning can improve temporal consistency through a lightweight modification to the pretrained model.

I introduce a bounding box LoRA module that uses only object information from the conditioning frame. The selected 5K checkpoint improves temporal consistency on both the checkpoint-selection and held-out test sets, with the held-out evaluation showing improvements across FID, FVD, and Agent Consistency. These results indicate that explicit object-aware conditioning can improve temporal stability through a lightweight modification to VISTA.

2. Experimental Objective

The baseline is the official pretrained VISTA model evaluated on a 300-sample nuScenes subset using 25-frame sequences conditioned on the first frame. The objective is to improve the temporal persistence of road agents across generated frames while keeping the modification small relative to the full model.

Only information available at inference time is used. Bounding boxes are taken exclusively from the conditioning frame, so no future-frame annotations or trajectories are provided. The final design adds an object-conditioning pathway to VISTA's temporal attention while leaving the rest of the generation pipeline unchanged.

3. Bounding Box Processing

For each nuScenes conditioning frame, 3D annotations are projected into the front-camera image to obtain 2D bounding boxes. Only annotations with visibility levels 3 and 4 are retained, which corresponds to 60%-100% visibility. The processed dataset contains 25,109 clips and 152,254 bounding boxes, with an average of 6.06 objects per clip. Of these clips, 1,622 contain no valid boxes, and the maximum number of boxes in a clip is 48.

The categories are car, truck, bus, trailer, construction vehicle, pedestrian, motorcycle, bicycle, traffic cone, and barrier. Each object is represented by its class and normalized center position, width, and height. Up to 32 objects are retained per sample; when more are present, the largest boxes by image area are selected.

4. Model Architecture

4.1 Bounding-Box Encoder

The bounding-box encoder combines semantic and geometric information for each object. The class label is mapped to a 64-dimensional learned embedding, while the normalized bounding-box coordinates are encoded by an MLP into a separate 64-dimensional feature. The two representations are concatenated and projected into a 256-dimensional object embedding.

4.2 Dense Spatial Representation

The object embeddings are converted into a dense spatial representation aligned with the latent feature map used by VISTA. Each 256-dimensional object embedding is assigned to the latent spatial locations covered by its bounding box. Features are averaged where boxes overlap, and an additional occupancy channel indicates whether each location belongs to an annotated object. The resulting representation therefore contains 257 channels at each latent spatial position: one occupancy value and a 256-dimensional object feature. This preserves locality by providing each temporal-attention location with object information specific to the corresponding image region.

4.3 Bounding Box Temporal Attention

The dense bounding box representation is injected into VISTA's temporal cross-attention layers. At each spatial position, the original temporal context contributes one key/value token, while the bounding box representation is projected into a second key/value token. These tokens are concatenated along the attention context dimension rather than added together, so the temporal attention layer explicitly attends over two separate sources: the original VISTA context and the bounding box context. This allows the model to learn how much attention to assign to each source when propagating information across the 25-frame sequence.

Bounding box information is applied only to temporal cross-attention. The spatial attention pathway is left unchanged because the goal is to improve persistence through time without globally altering spatial generation.

4.4 BBox-Specific LoRA

A separate rank-16 LoRA branch adapts the query, key, value, and output projections in the temporal cross-attention layers. Separate learned projections map the 257-dimensional local bounding box representation into additional attention key and value vectors. The final extension introduces approximately 8.2 million trainable parameters compared with roughly 2.5 billion parameters in the baseline VISTA model.

5. Evaluation Setup

Evaluation uses FID, FVD, and Agent Consistency, with FVD and Agent Consistency serving as the primary measures of temporal consistency. Agent Consistency is adopted from the DrivingGen evaluation framework, which detects road agents in the initial frame and tracks them across the generated sequence to measure object-level temporal stability.

Checkpoint selection is first performed on a fixed 300-sample evaluation set. BBox LoRA checkpoints at 3K, 5K, 7K, 10K, and 15K optimization steps are evaluated on the same samples to study the effect of training duration and select the best checkpoint. The selected checkpoint is then evaluated on a separate held-out set of 300 previously unseen samples. Baseline VISTA and the selected BBox LoRA model are tested on the same held-out samples using matched inference settings and native 320 × 576 resolution. This evaluation is used to test whether the observed improvement generalizes beyond the checkpoint-selection set.

6. Results

Checkpoint Selection Set:

Model	FID ↓	FVD ↓	Agent Consistency ↑
Real nuScenes	—	—	0.83
VISTA	**16.41**	306.74	0.69
BBox LoRA 3K	20.40	269.53	0.60
BBox LoRA 5K	20.06	**259.66**	0.71

BBox LoRA 7K	21.77	271.53	0.70
BBox LoRA 10K	20.30	267.16	0.70
BBox LoRA 15K	20.62	287.64	**0.72**

Held-Out Test Set:

Model	**FID ↓**	**FVD ↓**	**Agent Consistency ↑**
VISTA	23.23	335.93	0.70
BBox LoRA 5K	**20.75**	**268.07**	**0.71**

7. Discussion

The checkpoint selection results show a clear training duration effect. Performance improves substantially from 3K to 5K steps, with the 5K checkpoint achieving the lowest FVD of 259.66 while increasing Agent Consistency from 0.69 to 0.71 relative to baseline VISTA. Beyond 5K, additional training does not improve the overall balance of metrics. At 7K and 10K, FVD worsens while Agent Consistency remains around 0.70, and at 15K, Agent Consistency increases slightly to 0.72 but FVD deteriorates to 287.64. This suggests that longer training increasingly favors object persistence while sacrificing broader video quality.

The held-out test results provide stronger evidence that the improvement generalizes beyond the samples used for checkpoint selection. On 300 previously unseen samples, BBox LoRA 5K improves all three metrics relative to baseline VISTA. FID decreases from 23.23 to 20.75, a 10.7% improvement, while FVD decreases from 335.93 to 268.07, a 20.2% improvement. Agent Consistency also increases from 0.70 to 0.71.

The held-out result is especially important because the FID degradation observed on the checkpoint-selection set does not persist. This suggests that the earlier reduction in frame-level fidelity was not a consistent effect of the proposed method. Across both evaluation sets, the strongest and most consistent improvement is in FVD, while Agent Consistency shows a smaller but positive improvement at the selected checkpoint.

8. Conclusion

This experiment shows that bounding box conditioning can improve temporal consistency in VISTA through a lightweight LoRA-based modification. The proposed method encodes conditioning frame bounding boxes into spatially localized object features and injects them into temporal cross-attention through a bounding-box-specific LoRA pathway. Performance peaks around 5K training steps, while longer training leads to diminishing or negative returns in video quality. Overall, the experiment demonstrates that explicit object-level conditioning can improve the temporal stability of VISTA without requiring large-scale retraining.